

PASTTEL: Parallel analysis framework for Termination and non-Termination of Lasso programs

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Abstract. Proving termination or non-termination of *lasso programs* is a challenging problem in program verification. We present PASTTEL, a modular and generic parallel portfolio framework that unifies state-of-the-art (non-)termination analyses of lasso programs. PASTTEL (1) facilitates the integration of new analysis algorithms, (2) executes registered strategies concurrently, and (3) can be embedded as a self-contained C++ library into any project requiring (non-)termination analysis. To illustrate its genericity, we instantiated it with the strategies of ULTIMATE LASSORANKER by implementing the framework’s interfaces. On 5727 lassos, this instantiation reproduces the results of ULTIMATE LASSORANKER with the same verdicts and a lower analysis time, and it speeds up an existing verifier when plugged in as its first-line analyzer.

1 Introduction

Proving the termination of a program is one of the most fundamental correctness properties in computer science, and is known to be undecidable in the general case. A practical way to approach this problem is to search for explicit termination or non-termination arguments. The existence of a certain termination argument — such as a linear ranking function — is decidable [5,7,25,29,35] and implies termination. However, if no linear ranking function can be found, we cannot conclude non-termination. Conversely, non-termination arguments, e.g., recurrence sets [23] or geometric non-termination arguments [30], can be synthesized by constraint solving for restricted templates and imply non-termination, but a failure to find one cannot conclude termination.

These analyses are commonly applied to *lasso programs*: execution traces consisting of a single while loop preceded by straight-line code, whose control flow graph takes the characteristic shape of a lasso.

In typical use cases, a lasso program is extracted from a control flow graph to represent a candidate infinite execution path, as encountered in CEGAR-based

termination synthesis [12,13,14,18].

We introduce PASTTEL⁴, a generic and modular, parallel analysis framework for the termination and non-termination of lasso programs. PASTTEL unifies state-of-the-art termination techniques within a concurrent portfolio. Our main contributions are as follows:

- We propose a modular and generic framework for implementing termination and non-termination strategies, designed to facilitate their integration into a unified portfolio.
- We implement several state-of-the-art constraint-based techniques, including ranking function synthesis [29] and geometric non-termination arguments [30].
- We evaluate an instance of our framework on a benchmark from the SV-COMP 2025⁵, TERMCOMP 2025⁶ competitions, and the ULTIMATE⁷ repository. The results show that the modularity and genericity of PASTTEL do not compromise performance, while its C++-based implementation yields a significant speedup in execution time.

Beyond its standalone use, PASTTEL is designed as a self-contained library that any external project can integrate to leverage (non-)termination analysis capabilities without knowledge of its internal design.

The remainder of this paper is organized as follows. Section 2 surveys the state-of-the-art termination and non-termination analysis tools and positions PASTTEL among them. Section 3 introduces the formal definition of lasso programs. Section 4 describes the architecture of the framework and the modular orchestration of the analysis strategies. Section 5 showcases an instantiation of PASTTEL based on ULTIMATE LASSORANKER approaches. We conclude with an empirical evaluation of the tool in Section 6.

2 PASTTEL vs. the State-of-the-Art

To motivate the design of PASTTEL and highlight its originality, we first provide an overview of the state-of-the-art tools for (non-)termination synthesis.

Existing tools. Termination analysis is a very active field of research, and most tools operate at the whole-program level, iteratively extracting and analyzing individual lassos as part of their verification strategy. Among these, PROTON [32,33] integrates a fine-tuned LLM for ranking function synthesis, achieving a gold medal at SV-COMP2025⁵ in the Termination category. ULTIMATE

⁴ <https://github.com/Dowers/PaSTTeL/tree/artifact-vmcai27>

⁵ <https://sv-comp.sosy-lab.org/2025/>

⁶ <https://termination-portal.org/wiki/TPDB>

⁷ https://www.ultimate-pa.org/lasso_ranker/

AUTOMIZER [18,24], the overall winner of SV-COMP 2025, delegates termination to Ultimate Büchi Automizer, which implements a CEGAR loop combining ULTIMATE LASSORANKER with automata-theoretic techniques: termination arguments for individual lassos are generalized into Büchi automata representing all traces covered by the proof, which are then removed from the program via complementation and intersection. CPACHECKER [2] takes a different approach, extracting candidate lassos via predicate-based reachability analysis, but likewise delegates their (non-)termination analysis to ULTIMATE LASSORANKER. Beyond these, other whole-program tools such as APROVE [22,26], T2 [10], and VERY-MAX [6,27] also address termination through their own dedicated strategies.

While the tools above operate at the program level, a smaller subset focuses specifically on single-lasso analysis, which is directly relevant to our work. ULTIMATE LASSORANKER [25,28,29] is a Java library integrated into ULTIMATE AUTOMIZER verification framework⁸. It extracts and analyzes linear lasso programs then provides termination proofs based on ranking function templates [28,29] with linear inductive supporting invariants, and represents one of the state-of-the-art for single-lasso termination analysis over linear arithmetic. Similarly, IRANKFINDER [5] synthesizes linear, lexicographic-linear, and multiphase-linear ranking functions for multi-path loops. On the non-termination side, ULTIMATE LASSORANKER combines *fixpoint* checking with the synthesis of *Geometric Non-Termination Arguments* (GNTAs) [30], which represent an infinite execution as a sum of geometric series. LOAT (Loop Acceleration Tool) [20,21] follows a different approach: it relies on loop acceleration, which computes closed-form descriptions of arbitrarily many loop iterations, to derive witnesses of non-termination.

Positioning PASTTEL among existing tools. In the tools surveyed above, modularity is rarely a design goal: adding a new (non-)termination technique typically requires deep knowledge of the tool’s internals. Concurrent portfolio execution at the lasso level is also rare. APROVE comes closest, running three backends concurrently (T2-based recurrent sets, LOAT-based acceleration, and KOAT [31]), but it is not open-source, which limits its reuse and extension. ULTIMATE LASSORANKER on the other hand is open-source, but it applies its analyses sequentially, in a static order, one after another.

PASTTEL is designed to address these limitations. It is an open-source, generic, and modular C++ framework that also exposes a library interface, enabling any external project to embed and leverage (non-)termination analysis capabilities as a self-contained component. New strategies can be integrated by implementing a single interface, making it straightforward to enrich the portfolio. Unlike APROVE, the portfolio is fully extensible. Unlike LASSORANKER, all registered strategies are executed concurrently, so that the first conclusive result is immediately returned without waiting for the remaining strategies to complete. Beyond the approaches currently integrated, the modular design of PASTTEL makes it straightforward to incorporate any existing technique. As a

⁸ <https://www.ultimate-pa.org/automizer/>

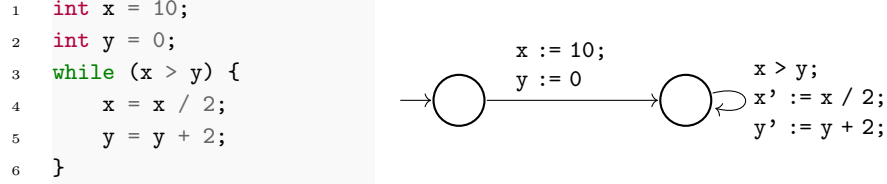


Fig. 1: The control flow graph of the left-side lasso program is defined by $\tau_{\text{stem}} = x := 10; y := 0$ and $\tau_{\text{loop}} = x > y; x' := x / 2; y' := y + 2;$

proof of concept, Section 5 presents an instantiation of PASTTEL based on the techniques underlying ULTIMATE LASSORANKER.

3 Lasso Program

A variety of techniques have been proposed for reasoning about lasso programs, including constraint-based methods [1,5,7,8,11,25,30,35], abstract interpretation [15,36], and algebraic approaches based on eigenvalue analysis [9]. As this work focuses primarily on constraint-based methods, we now introduce the formal definitions of (*linear*) lasso programs and the notion of termination considered throughout the paper.

We denote by $\mathbf{x} = (x_1, \dots, x_n)^\top \in \mathbb{R}^n$ the vector of n program variables, and by $\mathbf{x}' = (x'_1, \dots, x'_n)^\top \in \mathbb{R}^n$ the variables of the next state.

To abstract away from program syntax, we define a lasso program directly by the two relations that generate its execution sequences.

Definition 1 (Lasso Program [25,30]). *Given a set of states Σ , a lasso program $P = (\tau_{\text{stem}}, \tau_{\text{loop}})$ consists of two relations $\tau_{\text{stem}} \subseteq \Sigma \times \Sigma$ and $\tau_{\text{loop}} \subseteq \Sigma \times \Sigma$, called the stem and the loop of P , respectively.*

An execution of P is a possibly infinite sequence of states $\sigma_0, \sigma_1, \dots$ such that $(\sigma_0, \sigma_1) \in \tau_{\text{stem}}$ and $(\sigma_i, \sigma_{i+1}) \in \tau_{\text{loop}}$ for all $i \geq 1$. The program P is terminating if it admits no infinite execution.

A loop program is the special case where $\tau_{\text{stem}} \equiv \text{true}$.

Linear relations. For $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$, we write $A\mathbf{x} \leq \mathbf{b}$ for the conjunction $\bigwedge_{j=1}^m \sum_{i=1}^n a_{ji}x_i \leq b_j$ of linear constraints. A *linear relation* is a relation $\tau = \{(\mathbf{x}, \mathbf{x}') \in \mathbb{R}^n \times \mathbb{R}^n \mid A(\mathbf{x}, \mathbf{x}')^\top \leq \mathbf{b}\}$ with $A \in \mathbb{R}^{m \times 2n}$.

$$\tau = \{(\mathbf{x}, \mathbf{x}') \in \mathbb{R}^n \times \mathbb{R}^n \mid A \begin{pmatrix} \mathbf{x} \\ \mathbf{x}' \end{pmatrix} \leq \mathbf{b}\}$$

Definition 2 (Linear Lasso Program [25,30]). *A linear lasso program is a pair $P = (\tau_{\text{stem}}, \tau_{\text{loop}})$ where $\tau_{\text{stem}}, \tau_{\text{loop}} \subseteq \mathbb{R}^n \times \mathbb{R}^n$ are linear relations over the current-state variables $\mathbf{x}$ and next-state variables $\mathbf{x}'$.*

Figure 1 illustrates these definitions: the stem encodes the initialization $x := 10; y := 0$ and the loop the guarded transition $x > y; x' := x / 2; y' := y + 2;$

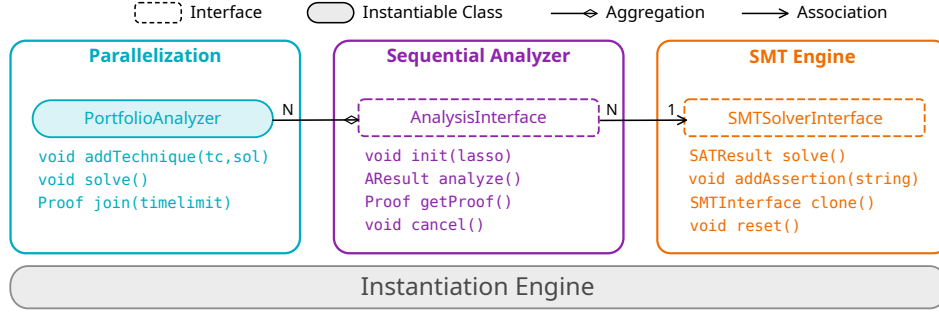


Fig. 2: The architecture of PASTTEL

Linearization of Lasso Programs

Some (non-)termination analysis strategies operate exclusively on linear lasso programs [25,28,29,30]. When the input lasso contains non-linear constructs outside linear arithmetic (Boolean variables, division, modulo, ...), a pre-processing linearization phase is initiated⁷. For instance, Boolean variables are replaced by integer comparisons over a fresh auxiliary variable v , where *true* is encoded as $v \geq 1$ and *false* as $v \leq 0$.

When full linearization is not achievable — for instance, when a product of two free variables $x \times y$ appears — three strategies may be adopted for the non-linear atom: it is replaced by $\top$ (over-approximation), by $\perp$ (under-approximation), or an *exception* is raised. Applied to the atoms of a negation-free formula, over-approximation keeps all behaviors of the original relation, so termination proofs remain valid whereas non-termination arguments may be spurious; dually, under-approximation removes behaviors, so non-termination arguments remain valid whereas termination proofs do not. All three strategies are implemented in PASTTEL and selected via a configuration parameter.

4 Architecture of the Framework

The performance of existing termination and non-termination analysis strategies heavily depends on implementation choices such as the programming language or the libraries used (*e.g.*, C++ vs. JAVA). This makes it difficult to compare strategies fairly without introducing a technological bias, or at best, fully reimplementing them to eliminate it. Moreover, integrating new strategies on top of existing ones remains a non-trivial task.

PASTTEL is designed to address these issues. The architecture of PASTTEL is structured around three core components: a sequential analyzer (**AnalysisInterface**); an SMT engine (**SMTSolverInterface**); and a parallelization layer (**PortfolioAnalyzer**), as illustrated in Figure 2). Any instantiation of the framework must provide concrete implementations of these three components.

SMT Engine: SMTSolverInterface. This adapter defines the basic functions expected from an SMT solver, allowing different solvers to be plugged in

interchangeably. PASTTeL currently supports Z3 [17] and CVC5 [3]. It is used by constraint-based strategies, which reduce the synthesis of a (non-)termination argument to a satisfiability problem. Each sequential analyzer holds its own solver instance, which it passes to its constraint-based strategies. The main methods of this interface are:

- `void addAssertion(string assertion)`: adds a constraint to the solver.
- `SATResult solve()`: checks the satisfiability of the current constraints.

Sequential Analyzer: AnalysisInterface. This adapter defines the basic functions expected from a sequential (non-)termination analyzer. The `LassoProgram` data structure used as input represents the lasso in linear format, with the stem and loop transitions, the list of program variables, and other information required for the analysis. The main methods of this interface are:

- `void init(LassoProgram* lasso)`: initializes the analyzer with the input lasso program and apply (if needed) preprocessing step as linearization.
- `AnalysisResult analyze()`: launches the analysis using the analyzer’s own SMT solver instance. Returns `TERMINATING`, `NON_TERMINATING`, or `UNKNOWN`.
- `ProofCertificate getProof()`: returns a proof certificate for the (non-)termination result in plain text or JSON format.
- `void cancel()`: interrupts the current search initiated by the caller.

Parallelization: PortfolioAnalyzer. This component orchestrates the concurrent execution of termination and non-termination strategies on the same input lasso. Strategies share no state: each one owns a private solver instance (for Z3, a dedicated `z3::context`), so that strategies are written as sequential code and no locking is needed on the solving path. The main methods of this component are:

- `void addTechnique(AnalysisInterface* technique, SMTSolverInterface* solver)`: registers a strategy with its private solver instance.
- `void solve()`: runs the registered strategies, in registration order, on a pool of $\min(n, |strategies|)$ threads, where n is the configured pool size. With $n = 1$, execution is strictly sequential, which yields the sequential configuration of Section 6 without any code change.
- `ProofCertificate join(int timelimit)`: blocks until a strategy returns a conclusive result or the time limit expires. The first conclusive verdict wins and its certificate is returned; since sound strategies cannot contradict each other, no arbitration is needed. All other strategies are then cancelled cooperatively, through a flag checked between template configurations and by interrupting their solver (e.g., via `Z3_interrupt`), as forced thread cancellation is unsafe inside solver libraries. On timeout, all strategies are cancelled and `UNKNOWN` is returned.

Engine Instantiation: init-PaSTTeL. To create a particular instance of PASTTeL, the user has to adapt the `init-PaSTTeL` function (Algorithm 1), which serves as the entry point for instantiating and binding all components of the framework. Its parameters expose the key configuration points, allowing any

Algorithm 1: init-PaSTTeL function

Input: *lasso*: linear lasso program, *n*: number of threads, *solverType*: {Z3, CVC5}

- 1 *solvers* $\leftarrow$ Create *n* of *solverType* from *SMTSolverInterface*;
- 2 *portfolio* $\leftarrow$ Create *PortfolioAnalyzer*(*lasso*, *n*);
- 3 *portfolio.addTechnique*(new *TerminationSA*(...), *solvers*[0]);
- 4 *portfolio.addTechnique*(new *NonTerminationSA*(...), *solvers*[1]);
- 5 *portfolio.addTechnique*(...);
- 6 *portfolio.solve*();
- 7 *proof* $\leftarrow$ *portfolio.join*(*timelimit*);
- 8 *print*(*proof*);

external tool to embed PASTTEL as a self-contained library with the desired configuration.

First, the *n* concrete solver instances of type *solverType* (inheriting from *SMTSolverInterface*) are instantiated (line 1). The portfolio analyzer is then instantiated with a reference to the *lasso* structure (line 2). Analysis strategies (inherited from *AnalysisInterface*) are registered into the portfolio (lines 3–5), each linked to a dedicated solver instance. The concurrent search is initiated via *solve* (line 6), and the main thread blocks until a conclusive result is found or the time limit is reached (line 7). The final result is retrieved, along with the proof certificate if one was produced (line 8).

5 PASTTEL "à la ULTIMATE LASSORANKER"

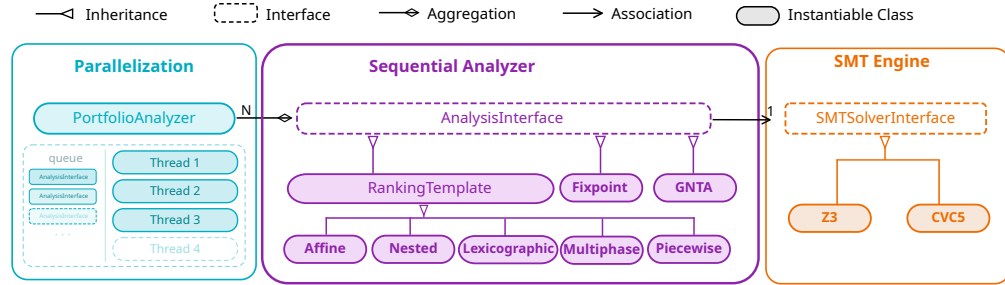


Fig. 3: P-ULR instantiation architecture

As a proof of concept to validate the generic aspect of our framework, we selected the techniques underlying ULTIMATE LASSORANKER (ULR), the lasso-level analysis library of ULTIMATE AUTOMIZER, winner of the Overall ranking of SV-COMP 2025. We implemented P-ULR, an instantiation of PASTTEL that replicates the core (non-)termination strategies of ULR. Each technique is realized as an independent analyzer registered into the portfolio and executed concurrently with the others. The resulting instantiation is illustrated in Figure 3.

```

1 {
2   "program_vars": [
3     {
4       "x": "Int", "y": "Int"
5     },
6   "stem": [
7     {
8       "formula": "(and (= x_0 10) (= y_0 0))",
9       "in_vars": {},
10      "out_vars": { "x": "x_0", "y": "y_0" }
11    },
12    "loop": [
13      {
14        "formula": "(and (> x_1 y_1) (= x_2 (div x_1 2)) (= y_2 (+ y_1 2)))",
15        "in_vars": { "x": "x_1", "y": "y_1" },
16        "out_vars": { "x": "x_2", "y": "y_2" }
17      }
18    ]
19  }

```

Fig. 4: Input format of lasso program in Figure 1

Input Format & Linearization. While PASTTEL operates directly on linear lasso program structures (see Section 4), P-ULR provides a JSON-based input format (see Figure 4) as a convenience layer, designed to be generic and tool-agnostic. Formulas are encoded following the SMT-LIB standard [4], with support for Single Static Assignment (SSA) form [16]. Each input file specifies the program variables and their types, together with the input and output variables of both τ_{stem} and τ_{loop} . Support for additional fields, including axioms and uninterpreted functions, is currently under development.

P-ULR consists of approximately 10,000 lines of C++ code, 20% of which implement the linearization preprocessing, built on top of a core framework of about 3,000 lines. Given a lasso program in the JSON-based SMT-LIB format, P-ULR first linearizes it when necessary, using a preprocessing phase that closely follows that of ULR, and then dispatches the analysis to the registered strategies. Since all implemented components inherit from the framework’s abstract interfaces, they are fully reusable and can be extended or complemented with additional techniques, such as new ranking function templates, without modifying the existing codebase.

Termination. P-ULR provides a constraint-based synthesis of termination arguments for linear lasso programs using linear ranking templates [28,29]. This approach generalizes constraint-based termination analysis by providing a unified framework for synthesizing ranking functions with affine-linear components, encompassing *nested*, *piecewise*, *parallel*, and *lexicographic* ranking functions, which can further be combined to form a more powerful composite templates. Following [29], the approach relies on Motzkin’s transposition theorem [34], a generalization of Farkas’ lemma [19] that handles both strict and non-strict inequalities, to transform the generated $\exists\forall$ -constraint into an $\exists$ -constraint directly solvable by an SMT solver. For instance, on the example of Figure 1, the synthesized ranking function is $f(x, y) = 2x - 2y - 1$, with supporting invariant $y + 2 \geq 0$. Currently, P-ULR implements *affine*, *nested*, *lexicographic*, *multiphase*, and *piecewise* ranking templates, which can be run concurrently, whereas ULR tries them

sequentially in a fixed order.

Non-Termination. P-ULR implements a constraint-based synthesis of non-termination arguments for linear lasso programs, integrating two approaches: *fixpoint* checking and *Geometric Non-Termination Arguments (GNTA)* [30]. A *GNTA* represents an infinite execution as a pointwise sum of geometric series: for $k \geq 2$, the k -th state is $\mathbf{x}_1 + \sum_{i=0}^{k-2} \lambda^i \cdot \mathbf{y}$, where $\mathbf{x}_1$ is the program state before the first loop iteration, $\mathbf{y}$ is the initial direction of the execution, and λ is the rate at which the execution keeps moving in that direction. This approach is complete for deterministic lassos with non-negative eigenvalues and reduces the search to an existential constraint in non-linear integer or real arithmetic (depending on the variable types), which can be discharged by an SMT solver. *Fixpoint* checking is the special case of *GNTA* where $\mathbf{y} = \mathbf{0}$: it searches for a state $\mathbf{x}$ such that $(\mathbf{x}, \mathbf{x}) \in \tau_{\text{loop}}$, i.e., a state from which the loop can iterate indefinitely without changing. As described in Section 2, ULR applies these two methods sequentially; P-ULR instead runs them concurrently, so that the *GNTA* analysis does not have to wait for the outcome of *fixpoint* checking.

Limitations. The current version of P-ULR has several limitations. First, the linearization step does not yet cover all non-linear cases: in particular, array operations such as constant arrays, array-valued `ite`, and array disequalities, as well as uninterpreted functions with quantified axioms, are not completely supported. Second, at the framework level, the `LassoProgram` structure currently provides better support for linear formulas; non-linear formulas are passed as-is to the registered strategies, which is valid for strategies that do not require linearization such as *fixpoint*. Third, at the strategy level, P-ULR currently supports linear rational arithmetic (QF_LRA) for termination synthesis and linear integer arithmetic (QF_LIA) for *GNTA* synthesis. Finally, simplification of synthesized ranking function coefficients and formula optimization have not yet been implemented.

6 Experimental Evaluation

Our evaluation does not aim to outperform ULR, but to assess the framework itself, using ULR as a reference whose techniques and results are well established. We address three questions: **(RQ1)** Can PASTTEL host existing techniques faithfully, i.e., does its instantiation return the same verdicts as ULR? **(RQ2)** Does the portfolio layer bring benefits on top of the hosted strategies, without modifying them? **(RQ3)** Can a PASTTEL instantiation be embedded into an existing verifier with little effort?

Experimental Setup. The experiments were conducted on a machine with 16 vCPUs of an Intel Core i5-14400T processor and 30 GiB of memory. We used 2856 programs: 1557 C programs from the SV-COMP 2025⁵ and TERMCOMP

Tool	#Terminating	Time (s)	#Non-Terminating	Time (s)
ULR-BASELINE		1173		20
P-ULR-SEQ	4141	102	1586	3
P-ULR-PAR7		47		3

Table 1: Comparison of P-ULR with ULR on **5727** lasso programs. Times are accumulated wall-clock times over commonly solved lassos.

2025⁶ competitions, and 1299 Boogie programs from the ULTIMATE⁸ repository.

We use two time-outs: 3000 s per program for the whole-program verification runs used to build the benchmark, so as to collect as many lassos as possible, and 600 s per lasso for lasso-level analyses. To build the benchmark, we ran ULTIMATE BÜCHI AUTOMIZER, which relies on ULR for lasso-level analysis, on each program and serialized to JSON every lasso submitted to ULR, yielding 5727 lassos. The time spent by ULR on each strategy was logged in the corresponding lasso file, and its total time on a lasso never exceeded 600 s. P-ULR was then evaluated on the same lassos under the per-lasso time-out.

RQ1: Faithfulness of the instantiation. To compare P-ULR and ULR under the same conditions, we disabled the parallel execution of P-ULR, yielding P-ULR-SEQ, and compared it with ULR, referred to as ULR-BASELINE. ULR-BASELINE executes its strategies in a fixed order: *fixpoint* checking then *GNTA* for non-termination, and the *affine*, *nested*, *multiphase*, and *lexicographic* templates for termination. P-ULR-SEQ follows the same order, with the *piecewise* template as an additional last strategy. Both tools use Z3 (v4.16.0) as SMT solver.

The first two rows of Table 1 report, for each tool, the number of lassos proved terminating and non-terminating, and the accumulated solving time over the commonly solved instances. On all 5727 lassos, P-ULR-SEQ and ULR-BASELINE return identical verdicts. This answers RQ1 positively: the techniques of ULR can be hosted in PASTTEL with identical results. P-ULR-SEQ is also faster, with a total gain of 1088 s (~ 18 min); in particular, the accumulated time for termination synthesis drops from 1173 s to 102 s (a factor of 11). Since this gain stems from the implementation and from the absence of ULR’s partitioning step (Section 5), we read it as an upper bound rather than an algorithmic improvement.

The scatter plots in Figure 5 show the same results on a logarithmic time scale.

- On terminating instances, P-ULR-SEQ is faster than ULR-BASELINE on most lassos. This gain should be put into perspective, as most of the benchmark lassos admit an *affine* ranking function, so that the more expressive templates are rarely needed. The synthesized ranking functions may also differ, since ULR-BASELINE relies on optimizations not yet integrated in P-ULR; in particular, without the partitioning step, P-ULR produces ranking functions with larger coefficients.

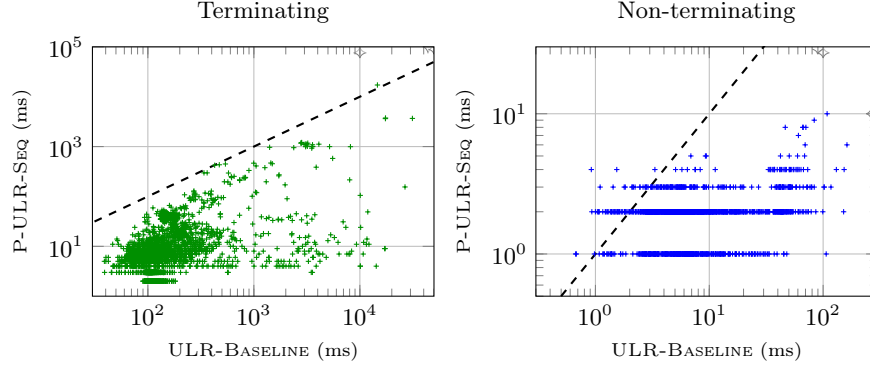


Fig. 5: Scatter plots of P-ULR-SEQ against ULR-BASELINE

- On non-terminating instances, two clusters appear: instances solved by *fixpoint* checking (bottom left), where both tools perform similarly, and instances requiring a *GNTA* (upper right), where P-ULR-SEQ is clearly faster.

RQ2: Benefit of the portfolio layer. We then run P-ULR in its concurrent mode with 7 threads, referred to as P-ULR-PAR7. Each thread runs an independent `SequentialAnalyzer` instantiated with one of the following strategies: *fixpoint*, *GNTA*, and the *affine*, *nested*, *lexicographic*, *multiphase*, and *piecewise* ranking templates. The last four are parameterized by the number n of affine-linear functions (phases, components, or pieces) that jointly form the termination argument; since they reduce to the *affine* template for $n = 1$, their threads iterate over $n = 2, 3, 4$. The strategies themselves are identical to those of P-ULR-SEQ: only their orchestration differs.

The results are reported in the third row of Table 1. P-ULR-PAR7 reduces the total execution time by a factor of 23, achieving a gain of 1143 seconds (~ 19 min) over ULR-BASELINE. However, compared to P-ULR-SEQ, the parallel execution yields little additional speedup. This is explained by the composition of the benchmark: 80% of lasso programs admit an *affine* ranking function, which P-ULR-SEQ already resolves first in its sequential order, leaving limited opportunity for parallelism to bring further improvement on this benchmark.

RQ3: Embedding into an existing verifier. To assess a PASTTEL instantiation within a complete verification pipeline, we added to ULTIMATE BÜCHI AUTOMIZER a plugin of 520 lines of code that invokes P-ULR as an external (non-)termination analyzer. P-ULR acts as a first-line analyzer: it runs its strategies concurrently on each lasso, and the analysis falls back to LASSO-RANKER whenever it fails, so that the capabilities of LASSO-RANKER remain available.

We compared this configuration, ULTIMATE-PL, with the default one, ULTIMATE-LR, on 1087 C and Boogie programs from our benchmark (Figure 6), with a time-out of 1000 s per program, close to the 15-minute CPU-time limit used in

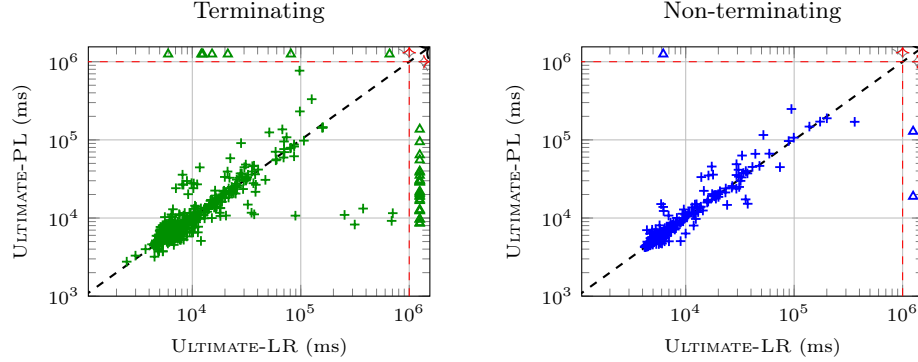


Fig. 6: Comparison of ULTIMATE-PL with ULTIMATE-LR on terminating (left) and non-terminating (right) programs. Programs solved by both tools are marked with +, those solved by only one tool with Δ , the failing tool being placed beyond the timeout lines (red).

SV-COMP. On the 883 programs solved by both, ULTIMATE-PL reduces the cumulative verification time from 13,491 s to 12,763 s (-5.4%). This end-to-end gain is more modest than in isolation, since lasso analysis accounts for only part of the verification time. ULTIMATE-PL also solves 28 programs on which ULTIMATE-LR fails, while failing on 8 programs solved by ULTIMATE-LR, due to time-outs or unsupported operations of the ULTIMATE pipeline. This answers RQ3 positively: a PASTTEL instantiation can be embedded into an existing verifier with a small integration effort, and benefits it.

7 Conclusion

We presented PASTTEL, a generic and modular framework for termination and non-termination analysis of linear lasso programs. By decoupling analysis strategies, SMT solvers, and their orchestration through abstract interfaces, PASTTEL makes it straightforward to integrate new techniques, to run them as a concurrent portfolio, and to embed the resulting analyzer as a library into any verifier. We illustrated this genericity by instantiating the framework with the techniques of ULTIMATE LASSORANKER: the resulting analyzer reduces the analysis time of lasso programs and, once plugged into ULTIMATE, speeds up end-to-end verification.

Several directions are planned as future work. First, we intend to extend the linearization pre-processing step to a broader class of programs. Second, we plan to enrich the portfolio with new strategies, such as recurrence set synthesis for non-termination, and to evaluate PASTTEL on benchmarks requiring more diverse termination arguments. Finally, the modular architecture opens the way to cooperation between strategies, such as sharing supporting invariants or lemmas, or stopping an analysis early based on information gathered by others, turning the portfolio into a truly collaborative analysis.

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